Evaluation of Phase Change Materials for Thermal Energy Storage in Wildland Firefighter Protective Shelters

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Objectives

The objective of this project is to improve existing emergency fire shelters by incorporating phase-change materials (PCMs) to reduce heat transfer to the interior of the shelter and extend the time of thermal protection for firefighters.

Introduction

Over the past 20 years, wildfires have increased in size, acreage burned, and instensity. In 2026, **112.3 million hectares** burned globally [1], and the U.S. firefighter entrapment incidents reached a **20 year high** [2].

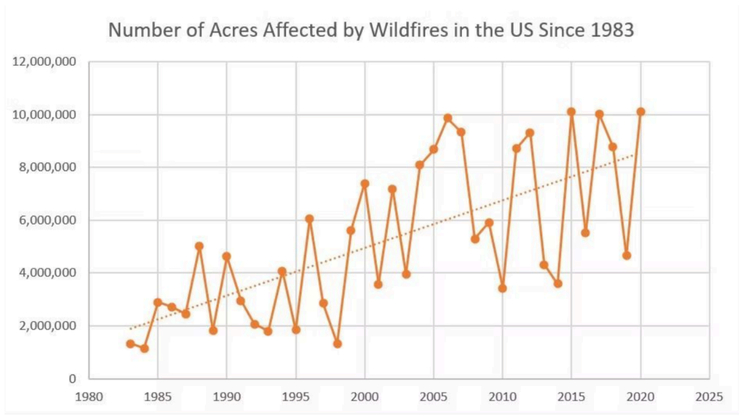


Figure 1: Annual U.S. wildfire acreage from 1983-2020.

Yarnell Hill Fire (2013)

A wildfire near Yarnell, Arizona killed 19 members of the Granite Mountain Hotshots on June 30 despite deploying emergency fire shelters [5].

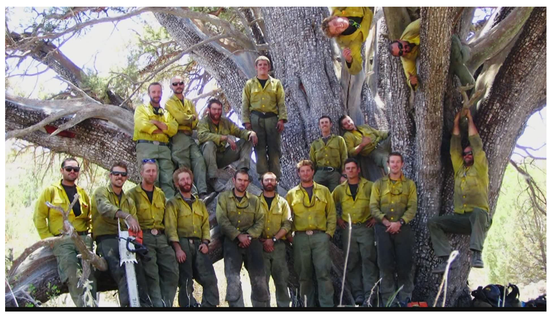


Figure 2: Granite Mountain Hotshots.

Emergency Fire Shelters

Emergency fire shelters provide last-resort protection to wildland firefighters when trapped by wildfires. They protect against radiant and convective heat but have limited effectiveness under extreme conditions.



Figure 3: Emergency fire shelter deployment procedure.

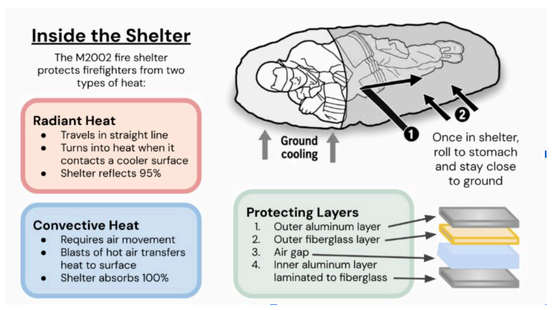


Figure 4: Fire shelter thermal protection mechanisms and material layers.

Methodology

This study compared and evaluated three organic and inorganic PCM layers in existing fire shelters, M-2002 fire shelter:

- RT42 Paraffin (Organic)
- PlusICE S46 (Inorganic)
- SP50 Gel (Inorganic)

Property	RT42	S46	SP50 Gel
Type	Organic	Salt Hydrate	Inorganic
Melting Temp. (°C)	42	46	49–50
Latent Heat (kJ/kg)	165	110	220
Solid Density (kg/m^3)	830	1610	1500
Liquid Density (kg/m^3)	760	–	1400
Conductivity (W/mK)	0.20	0.60	0.60

Table 1: Material Properties of Selected PCMs.

Layer transient analysis was performed in MATLAB using the ode45 function. A 6-node model represents the current M-2002 fire shelter, and the 7-node model incorporates an additional PCM layer. The layers were tested against a fire temperature of 1200 C for 30 minutes.

Transient Heat Model

The baseline M-2002 shelter was modeled using 6 nodes, and the PCM-integrated shelter was modeled using 7 nodes. Each layer is connected by thermal resistances and solved in MATLAB.

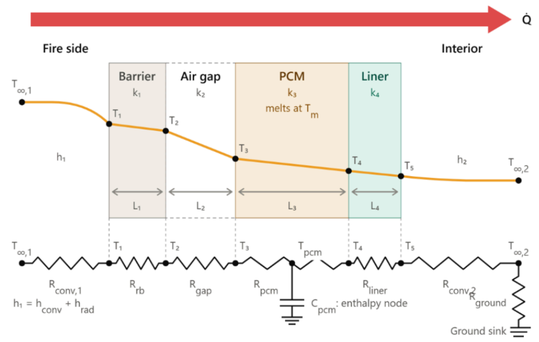


Figure 5: Thermal Resistance Network.

Governing Equations

Energy Balance:

$$m_i c_i \frac{dT_i}{dt} = Q_{in,i} - Q_{out,i}$$

PCM Node:

$$\frac{dE_{PCM_5}}{dt} = Q_{45} - Q_{56}$$

$$T_5 = T_{PCM,mp} + \frac{E_{PCM} - Q_{latent}}{mc_{p5}}$$

Total Shelter Resistance:

$$R_{total} = \frac{1}{h_1 A} + \sum_i \frac{L_i}{k_i A} + \frac{1}{h_2 A}$$

Heat Sink:

$$\frac{dT_7}{dt} = \frac{Q_{67} - Q_{ground}}{mc_{p7}}$$

Results

The current M-2002 shleter reached the interior temperature of 122 C in 11.6 seconds.

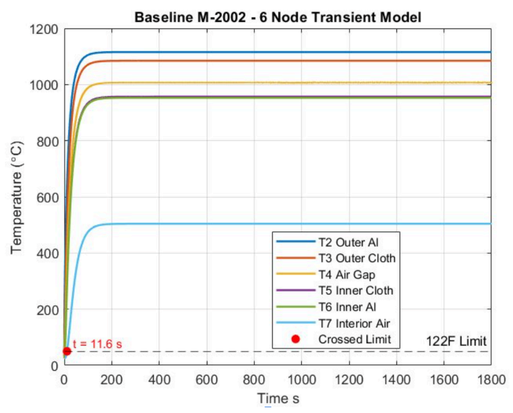


Figure 6: Baseline M-2002 - 6 Node Transient Model.

Results

PCM materials were evaluated using 7 node transient models.

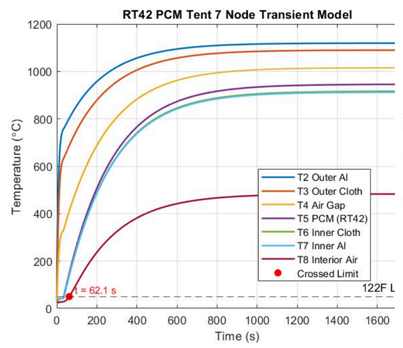


Figure 7: RT42 Paraffin.

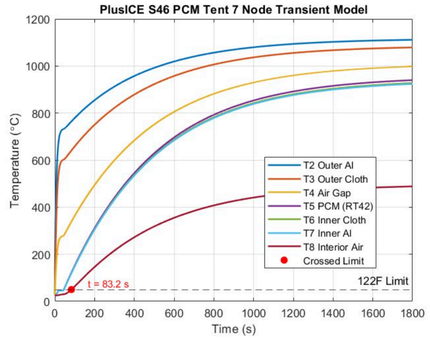


Figure 8: PlusICE S46.

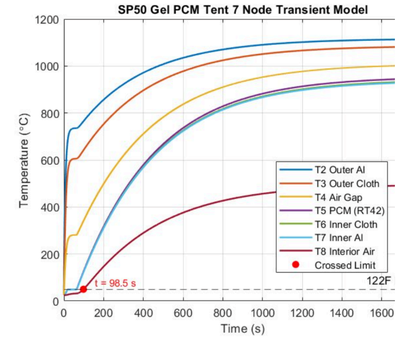


Figure 9: SP60 Gel.

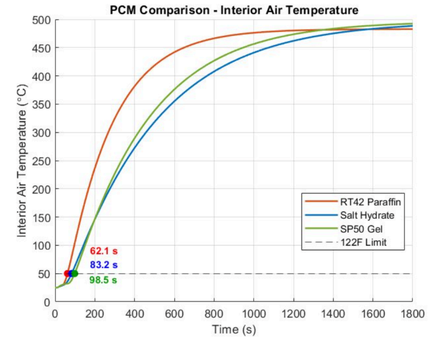


Figure 10:PCM comparison.

Material	Time (s)	Additional Mass (kg)	Improvement
Baseline	11.6	–	–
RT42 Paraffin	62.1	4.663	5.4×
PlusICE S46	83.2	9.046	7.2×
SP50 Gel	98.5	8.428	8.5×

Table 2: Time to reach 122 C and additional mass.

Adding the PCMs SP50 Gel, RT42 Paraffin, and Plus ICE S46 increased the heat transfer time to reach the interior temperature limit fo 122 C.SP 50 Gel increased the time the longest, and RT 42 Paraffin added the least amount of extra mass.

Conclusions

All three materials increased the time to reach the interior temperature of the shelter to the limit.

- **RT42 Paraffin:** Reached 122 C in 62.1 seconds with an added mass of 4.663 kg.
- **PlusICE S46:** Reached 122 c in 83.2 seconds with an added mass of 9.046 kg.
- **SP50 Gel:** Reached 122 C in 98.5 seconds with an added mass of 8.428 kg.

SP50 Gel increased the protection time the most, and RT42 Paraffin required the least amount of additional weight.

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Acknowledgements

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